FINAL REPORT

EdTech Student Performance and Engagement Portal

Track: Education Data Systems

Prepared By

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Certificate

This is to certify that the Summer Internship Project entitled **"EdTech Student Performance and Engagement Portal"** submitted by **Mr. Rajaram Maremalla** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering** has been successfully carried out during the Summer Internship at **Akshar AI Pvt. Ltd.**

The project work was completed under the guidance and supervision of the concerned mentors and demonstrates the student's ability to apply theoretical knowledge to practical problems in the field of Data Analytics and Educational Data Systems.

To the best of our knowledge, the work presented in this report is original and has not been submitted, either in part or in full, to any other University or Institution for the award of any degree, diploma, or certificate.

Guide Signature

Head of Department

Declaration

I hereby declare that the Summer Internship Project Report entitled **"EdTech Student Performance and Engagement Portal"** submitted by me in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** is a record of original work carried out by me during my internship at **Akshar AI Pvt. Ltd.**

The work presented in this report has been completed under the guidance of the concerned faculty and industry mentors. The project involved the study, analysis, and implementation of data analytics techniques to evaluate student performance and engagement using educational datasets. The findings, interpretations, and conclusions presented in this report are based on the work carried out during the internship period.

I further declare that this project report has not been submitted, either in part or in full, to any other University, Institution, or Organization for the award of any degree, diploma, certificate, or other academic qualification.

I also affirm that all sources of information used in this project have been appropriately acknowledged and referenced wherever necessary. Any contribution made by other individuals, organizations, or published resources has been duly recognized in the report.

I understand that any violation of the above declaration may result in disciplinary action as per the rules and regulations of the University.

Rajaram Maremalla

Enrollment No.: ____________________

Date: _______________

Place: _______________

Student Signature

Acknowledgement

It gives me immense pleasure to present this project report entitled **"EdTech Student Performance and Engagement Portal"**, which was successfully completed during my Summer Internship. The completion of this project would not have been possible without the support, guidance, and encouragement of several individuals and organizations to whom I express my sincere gratitude.

First and foremost, I would like to express my heartfelt thanks to **AKshar AI Pvt. Ltd.** for providing me with the opportunity to undertake my internship in a professional and learning-oriented environment. The internship provided valuable exposure to real-world industry practices, data analytics methodologies, and project development processes that greatly enhanced my technical and professional skills.

I extend my sincere gratitude to my Industry Mentor for their continuous guidance, encouragement, and constructive suggestions throughout the internship period. Their expertise, valuable feedback, and practical insights helped me understand the project requirements and successfully complete the assigned tasks.

I would also like to express my profound gratitude to my Faculty Guide, whose constant support, motivation, and academic guidance played a significant role in the successful completion of this project. Their suggestions and encouragement helped me overcome various challenges encountered during the development and documentation phases of the project.

I sincerely thank everyone who contributed in any way towards the successful completion of this internship project and report.

Rajaram Maremalla

Enrollment No.: ____________________

Abstract

Educational institutions generate large amounts of student-related data through examinations, assignments, attendance systems, and learning activities. Extracting meaningful insights from such data helps educators understand factors influencing student performance.

This project analyzes the Student Performance Dataset consisting of 649 student records and 33 attributes. The study investigates the impact of demographic factors, family background, study habits, previous failures, internet access, and educational support on academic achievement.

Various data preprocessing techniques, exploratory data analysis methods, statistical evaluations, and visualization techniques have been applied to identify meaningful patterns and trends.

The findings obtained from this analysis can assist educational institutions in improving student outcomes, identifying at-risk students, and supporting data-driven academic decisions.

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1 Introduction

1.1 Project Overview

The Student Performance Analysis project focuses on understanding the various factors that influence academic achievement among students. Educational institutions generate a large amount of data through examinations, assignments, attendance records, assessments, and student activities. However, much of this valuable information remains underutilized because it is often stored in separate systems and not analyzed systematically. This project aims to transform raw educational data into meaningful insights that can support better academic planning and decision-making.

The project utilizes the Student Performance Dataset, which contains detailed information about students' demographic characteristics, family background, educational support, study habits, social behavior, and academic results. By analyzing these attributes, it becomes possible to identify patterns and trends that contribute to student success or academic difficulties.

Data analytics techniques have been applied to explore relationships among different variables and understand their impact on final grades. Visualization methods such as bar charts and pie charts are used to present findings in a simple and understandable manner. These visual representations help educators and administrators quickly interpret important trends.

The ultimate goal of this project is to support data-driven educational practices. The insights obtained from the analysis can help institutions identify students who may require additional support, improve academic policies, and enhance overall educational outcomes.

1.2 Background of the Study

Educational Data Mining (EDM) is an emerging interdisciplinary field that combines data science, statistics, machine learning, and educational research. The primary objective of EDM is to discover meaningful patterns within educational datasets and use those insights to improve teaching and learning processes. As educational institutions increasingly adopt digital systems, large amounts of student-related data become available for analysis.

Student performance prediction has become one of the most important applications of educational data mining. Researchers and educators seek to understand the factors that contribute to academic success and identify students who may be at risk of poor performance. Such information can help institutions implement timely interventions and provide personalized academic support.

Previous studies have shown that academic achievement is influenced by numerous factors including study time, attendance, parental education, family support, access to technology, and socioeconomic conditions. Understanding the interaction between these variables enables educational institutions to design more effective learning environments and support systems.

With the growing importance of data-driven decision making in education, analytical approaches are becoming increasingly valuable. This project contributes to this field by examining student performance data and identifying the key factors that influence educational outcomes.

1.3 Problem Statement

Educational institutions face increasing challenges in managing and analyzing the large volumes of data generated through online learning platforms, including attendance records, assessment results, assignment submissions, and student engagement activities. Since this information is often distributed across multiple systems, obtaining a comprehensive view of student performance becomes difficult. Manual monitoring methods are time-consuming, prone to errors, and may not provide timely insights for effective academic intervention. Additionally, data quality issues such as duplicate records, missing values, and inconsistencies further complicate the analysis process, limiting the ability of educators and administrators to make informed decisions.

Another major challenge is the early identification of students who may be at risk of poor academic performance or declining engagement. Traditional evaluation methods often fail to provide real-time indicators, reducing opportunities for timely support and corrective action. Furthermore, institutions require secure and user-friendly systems that protect sensitive student information while presenting complex data through clear dashboards, reports, and performance indicators. The absence of a centralized analytics platform restricts effective educational monitoring and decision-making. Therefore, the proposed EdTech Student Performance and Engagement Portal aims to provide a unified, secure, and data-driven solution for collecting, analyzing, and visualizing educational data to improve student outcomes and institutional effectiveness.

1.4 Project Objectives

The primary objective of this project is to analyze student performance data and identify the factors that influence academic achievement. The project seeks to transform educational data into meaningful insights that can support students, educators, and academic administrators.

The specific objectives of the project are as follows:

- To study the characteristics of the Student Performance Dataset.
- To perform data cleaning and preprocessing operations.
- To conduct exploratory data analysis using statistical and graphical techniques.
- To identify relationships between demographic, social, and academic variables.
- To evaluate the influence of study habits, family support, and educational resources on academic performance.
- To generate visual representations of important trends and patterns.
- To identify factors associated with high and low academic achievement.
- To support educational decision-making through data-driven insights.
- To improve understanding of student learning behavior and performance indicators.

1.5 Scope of the Project

The scope of this project is focused on analyzing the Student Performance Dataset using data analytics techniques. The study covers data collection, data understanding, preprocessing, exploratory analysis, visualization, and interpretation of results. The project aims to identify patterns and relationships that contribute to student academic performance.

The analysis includes examination of demographic information, family characteristics, study habits, educational support systems, attendance records, and academic grades. Various statistical measures and graphical representations are used to understand the distribution of variables and their impact on educational outcomes.

The project is limited to the available dataset and does not include real-time student monitoring or live educational systems. Machine learning prediction models are outside the primary scope unless implemented as future enhancements. The findings are intended to provide insights that can assist educational institutions in improving student performance and academic planning.

1.6 Project Deliverables

The successful completion of this project results in several important deliverables that support both academic and practical objectives.

- Cleaned and validated Student Performance Dataset.
- Data preprocessing and transformation notebooks.
- Exploratory Data Analysis (EDA) reports.
- Statistical summaries and descriptive analysis.
- Graphical visualizations including bar charts and pie charts.
- Identification of key performance indicators and influencing factors.
- Comprehensive final project report.
- GitHub repository containing project files and documentation.
- Recommendations and insights for educational improvement.

These deliverables collectively demonstrate the complete data analytics workflow from raw data processing to final interpretation and reporting.

1.7 Report Organization

This report has been organized into seven chapters to provide a structured presentation of the project work. Each chapter focuses on a specific stage of the data analytics process and contributes to the overall understanding of student performance analysis.

Chapter 1 introduces the project, its objectives, scope, problem statement, and overall significance. Chapter 2 presents the requirement analysis, including domain study, existing systems, proposed solutions, and technologies used.

Chapter 3 discusses the dataset, data source, structure, data dictionary, and quality assessment. Chapter 4 explains the preprocessing activities performed to prepare the data for analysis.

Chapter 5 presents the exploratory data analysis, key metrics, findings, and relationships identified within the dataset. Chapter 6 discusses the results, interpretations, and limitations of the study.

Finally, Chapter 7 concludes the project by summarizing the findings and suggesting future enhancements and research opportunities.

2 Requirement Analysis

2.1 Domain Study

Educational Data Mining (EDM) is an interdisciplinary research field that combines education, statistics, data science, and machine learning to extract meaningful information from educational datasets. The primary objective of EDM is to improve learning outcomes by analyzing student behavior, academic performance, attendance records, and various educational indicators.

In modern educational institutions, a large amount of data is generated through examinations, assignments, attendance systems, online learning platforms, and student management systems. Traditionally, much of this information has been used only for record keeping. However, advancements in data analytics have enabled institutions to transform this raw data into actionable insights that support better academic planning and decision-making.

Student performance analysis is one of the most important applications of educational data mining. By studying factors such as study habits, family background, educational support, attendance, and social behavior, researchers can identify patterns that influence academic achievement. These insights help educators understand student needs and provide timely support to those who may be struggling academically.

Educational analytics also plays a vital role in identifying at-risk students. Through continuous monitoring and analysis of performance indicators, institutions can implement intervention programs before academic difficulties become severe. Such proactive approaches contribute to improved student retention, higher academic success rates, and enhanced educational quality.

The Student Performance Dataset used in this project provides a comprehensive view of student characteristics and academic outcomes. The dataset contains demographic, social, family, and educational attributes that make it suitable for educational research and performance analysis. Through systematic analysis of these variables, meaningful insights can be generated regarding the factors that contribute to student success.

Overall, educational data analytics supports evidence-based decision-making and helps institutions improve teaching strategies, optimize resource allocation, and enhance learning experiences for students.

2.2 Existing System Analysis

The existing system for evaluating student performance primarily relies on traditional assessment methods such as examinations, assignments, attendance records, and periodic evaluations. Teachers and administrators manually review these indicators to assess student progress and academic achievement.

Although traditional evaluation methods are widely used, they have several limitations. Most assessments focus only on examination results and often fail to consider the broader range of factors influencing academic performance. Important variables such as family support, study habits, social behavior, and access to educational resources may not receive sufficient attention during evaluation.

Another challenge of the existing system is the lack of comprehensive data analysis. Educational institutions collect large volumes of student-related information, but much of this data remains underutilized. Manual analysis methods are time-consuming and may not effectively identify hidden patterns or relationships between variables.

Traditional systems also face difficulties in identifying students who may be at risk of poor academic performance. Since interventions are often based on visible academic decline, support may be provided only after significant problems have already occurred. This reactive approach limits opportunities for early intervention and personalized assistance.

Furthermore, existing systems generally lack advanced visualization and reporting capabilities. Decision-makers often rely on summary reports that may not provide sufficient insight into student learning behavior or performance trends.

These limitations highlight the need for a more systematic and data-driven approach to student performance analysis.

2.3 Proposed System

The proposed system is a Student Performance Analysis framework based on educational data analytics. The system utilizes the Student Performance Dataset to investigate the factors that influence academic achievement and generate meaningful insights through statistical analysis and visualization techniques.

The proposed approach begins with data collection and understanding, followed by data preprocessing activities such as cleaning, validation, and transformation. Once the dataset has been prepared, exploratory data analysis is performed to identify important patterns, trends, and relationships among variables.

The system analyzes various student-related factors including demographic characteristics, family background, educational support, study habits, attendance records, and social activities. These factors are examined to determine their impact on academic performance.

Visualization techniques such as bar charts and pie charts are used to present findings in an understandable format. These visual representations help educators and administrators interpret analytical results quickly and effectively.

The proposed system also supports data-driven decision-making by identifying key performance indicators and factors associated with academic success. Educational institutions can use these insights to design targeted intervention strategies and improve learning outcomes.

Overall, the proposed system provides a structured and analytical approach to understanding student performance and supports evidence-based educational planning.

2.4 Functional Requirements

Functional requirements define the core operations that the system must perform to achieve its objectives.

- **Data Loading:** The system must be capable of importing and reading the Student Performance Dataset from CSV format for further analysis.
- **Data Cleaning:** The system should identify and handle inconsistencies, formatting issues, missing values, and duplicate records within the dataset.
- **Data Transformation:** The system should convert data into suitable formats for analysis and visualization.
- **Exploratory Data Analysis:** The system should perform descriptive and statistical analysis to understand dataset characteristics.
- **Visualization:** The system should generate charts and graphical representations to communicate findings effectively.
- **Relationship Analysis:** The system should identify relationships between different variables affecting student performance.
- **Statistical Analysis:** The system should calculate important metrics such as averages, distributions, frequencies, and correlations.
- **Report Generation:** The system should support preparation of analytical reports summarizing findings and conclusions.

2.5 Non Functional Requirements

Non-functional requirements define the quality attributes and operational characteristics of the system.

- **Reliability:** The system should produce consistent and dependable analytical results without errors.
- **Accuracy:** Data processing and analytical operations should provide accurate and valid results.
- **Scalability:** The system should be capable of handling larger educational datasets if required in future studies.
- **Security:** Student data should be handled responsibly and protected against unauthorized access.
- **Usability:** The system should present results in a simple and understandable manner for educators and administrators.
- **Maintainability:** The project should be organized and documented so that future modifications can be performed easily.
- **Performance:** Analytical operations should be completed efficiently without excessive computational delays.
- **Portability:** The project should be executable on different computing environments supporting Python and Jupyter Notebook.

2.6 Tools and Technologies Used

Several software tools and technologies were used throughout the project lifecycle to perform data analysis, visualization, documentation, and version control.

Table 2.1: Tools and Technologies Used

Tool/Technology	**Purpose**
Python	Data analysis and processing
Pandas	Data manipulation and preprocessing
NumPy	Numerical computations and mathematical operations
Matplotlib	Data visualization and graph generation
Jupyter Notebook	Development environment for analysis and experimentation
GitHub	Version control and project management
Overleaf	Report preparation and documentation
CSV Dataset	Data storage format used for analysis

These technologies collectively supported efficient implementation of the project and facilitated the complete data analytics workflow from data preparation to final report generation.

3 Data Collection and Understanding

The success of any data analysis project depends heavily on a thorough understanding of the dataset being used. Before performing any preprocessing or analysis, it is essential to examine the source, structure, quality, and characteristics of the available data. This chapter focuses on understanding the Student Performance Dataset and evaluating its suitability for educational data mining and analytics.

3.1 Data Source

The dataset used in this project is the Student Performance Dataset, which is widely utilized in educational research and learning analytics studies. The dataset contains information related to students' academic performance, demographic characteristics, family background, study habits, and social factors.

The dataset was originally developed for educational data mining research and provides valuable information that can be used to investigate the factors affecting student success. The data is available in CSV format, making it easy to load, process, and analyze using Python-based data science tools.

The primary objective of using this dataset is to identify relationships between student attributes and academic outcomes. Through systematic analysis, the dataset helps reveal patterns that can support educational institutions in improving student performance and designing effective intervention strategies.

3.2 Data Description

The Student Performance Dataset contains records of 649 students and includes 33 different attributes describing various aspects of student life and academic achievement.

The dataset combines demographic information, family characteristics, educational support indicators, study habits, social activities, and academic grades. Such a diverse collection of variables makes the dataset highly suitable for exploratory analysis and educational research.

The target outcome of interest is the final grade (G3), which represents the student's final academic performance. Other variables help explain how different factors contribute to this outcome.

Key characteristics of the dataset are summarized below:

Dataset Name	Student Performance Dataset heightNumber of Records
649 heightNumber of Attributes	33 heightData Format
CSV heightDomain	Education heightTarget Variable
G3 (Final Grade) height	

Table 3.1: Dataset Summary

3.3 Data Structure

The dataset consists of both categorical and numerical variables. Categorical variables describe characteristics such as gender, school type, family support, internet access, and extracurricular activities. Numerical variables include age, study time, number of failures, absences, and academic grades.

The dataset structure enables researchers to perform both descriptive and inferential analysis. Since the dataset includes multiple types of variables, it supports a wide range of analytical techniques including visualization, correlation analysis, statistical testing, and predictive modeling.

The data structure can be categorized into the following groups:

- Demographic Information
- Family Background Information
- Educational Support Variables
- Social and Lifestyle Factors
- Academic Performance Indicators
- Attendance and Study Habits

The presence of these diverse categories provides a comprehensive view of factors influencing student academic performance.

3.4 Data Dictionary

A data dictionary provides detailed descriptions of dataset variables and their meanings.

Attribute	Description
school	Student's school

sex	Gender of student
age	Age of student
address	Urban or rural residence
famsize	Family size
Pstatus	Parent cohabitation status
Medu	Mother's education level
Fedu	Father's education level
Mjob	Mother's occupation
Fjob	Father's occupation
reason	Reason for choosing school
guardian	Student guardian
traveltime	Travel time to school
studytime	Weekly study time
failures	Number of past failures
schoolsup	Extra educational support
famsup	Family educational support
paid	Extra paid classes
activities	Extracurricular activities
internet	Internet access at home
higher	Desire for higher education
romantic	Romantic relationship status
famrel	Family relationship quality
freetime	Free time after school
goout	Social outing frequency
Dalc	Workday alcohol consumption
Walc	Weekend alcohol consumption
health	Health condition
absences	Number of absences
G1	First period grade
G2	Second period grade
G3	Final grade

3.5 Data Quality Assessment

Data quality assessment is an important step that helps determine whether a dataset is suitable for analysis. Poor-quality data can lead to misleading conclusions and inaccurate results.

Several quality checks were performed on the dataset before analysis. These checks included examining missing values, duplicate records, data consistency, and variable validity.

The assessment revealed that the dataset is relatively clean and well-structured. No significant missing values were observed, and the dataset contained consistent formatting across attributes.

The quality assessment process included:

- Missing Value Detection
- Duplicate Record Identification
- Data Type Verification
- Consistency Checking
- Outlier Examination

The results indicated that the dataset is suitable for further preprocessing and exploratory analysis.

4 Data Preprocessing

Data preprocessing is one of the most important phases of the data analytics lifecycle. Raw datasets often contain inconsistencies, duplicate records, missing values, and formatting issues that can negatively impact analysis. The objective of preprocessing is to prepare the dataset for accurate and meaningful analysis.

4.1 Data Cleaning

Data cleaning involves identifying and correcting errors within the dataset. During this phase, the dataset was examined for inconsistencies, invalid entries, and formatting problems.

Several validation checks were performed to ensure data accuracy and consistency. The cleaning process helped improve overall dataset reliability and reduced the possibility of generating misleading analytical results.

4.2 Handling Missing Values

Missing values are one of the most common problems in real-world datasets. Missing information can reduce analytical accuracy and introduce bias into results.

The Student Performance Dataset was examined thoroughly for missing values across all attributes. The analysis revealed that the dataset contains no significant missing values.

Since missing values were not present, no imputation techniques such as mean substitution, median replacement, or mode filling were required.

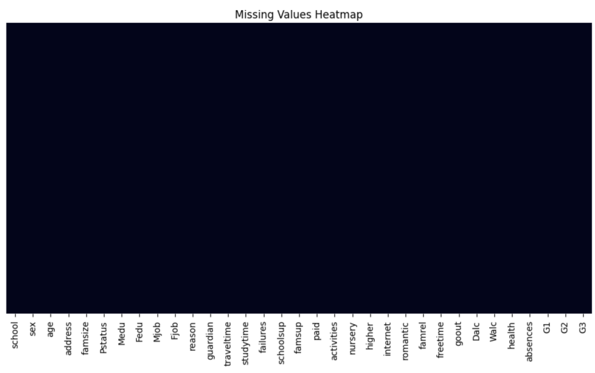

Figure 4.1: Handling Missing Values

4.3 Duplicate Removal

Duplicate records can distort statistical results and lead to inaccurate conclusions. Therefore, duplicate detection was performed during preprocessing.

The dataset was analyzed using duplicate record identification techniques. No significant duplicate records were found.

As a result, no records needed to be removed during this phase.

Figure 4.2: Duplicates

4.4 Data Transformation

Data transformation converts raw data into a format suitable for analysis. Several transformations were performed to improve analytical efficiency.

Categorical variables were encoded into machine-readable formats where necessary. Data types were verified and adjusted to ensure consistency across all attributes.

Transformation activities included:

- Categorical Variable Encoding
- Data Type Conversion
- Feature Formatting
- Variable Standardization

These transformations improved compatibility with analytical and visualization tools.

4.5 Data Validation

Data validation ensures that preprocessing activities have not introduced errors into the dataset.

Various validation checks were conducted after cleaning and transformation. These checks confirmed that all variables remained consistent and accurately represented their original meanings.

Validation activities included:

- Data Consistency Verification
- Record Count Validation
- Attribute Verification
- Data Type Confirmation

The validation process confirmed that the prepared dataset maintained its integrity and quality.

4.6 Final Prepared Dataset

After completing all preprocessing activities, the dataset was prepared for exploratory data analysis and visualization. Various data quality issues such as missing values, inconsistent records, and duplicate entries were carefully examined and addressed to ensure the reliability of the dataset.

The final dataset retained all valid records and contained clean, structured, and well-organized information suitable for educational analytics. Data cleaning procedures improved the overall quality of the dataset and minimized the possibility of errors affecting analytical results.

Each attribute was verified to ensure consistency and correctness. Categorical variables were standardized, numerical values were validated, and unnecessary redundancies were removed wherever required. The dataset was then transformed into a format that could be efficiently used for statistical analysis and machine learning tasks.

The prepared dataset provides a strong foundation for identifying student performance trends, investigating relationships between academic, social, and demographic factors, and generating meaningful educational insights. It also enables the creation of visualizations that support data-driven decision-making in educational environments.

Overall, the preprocessing phase successfully enhanced the dataset's quality, completeness, and usability, ensuring that subsequent analyses would produce accurate, reliable, and interpretable results.

5 Data Analysis and Modeling

This chapter presents the exploratory analysis conducted on the Student Performance Dataset. The objective of the analysis is to identify trends, patterns, and relationships that influence student academic performance.

5.1 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the dataset and identify important trends among students. Various graphical visualizations were created to examine demographic distributions, educational background, study habits, and academic outcomes.

5.1.1 School Distribution Analysis

The majority of students belong to the GP school, while a smaller proportion are from the MS school. This indicates that GP school contributes most of the records in the dataset and therefore has a stronger influence on overall trends.

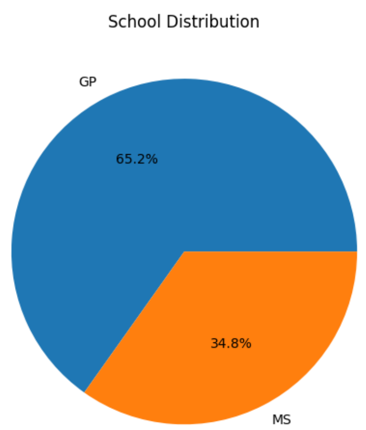

Figure 5.1: School Distribution

5.1.2 Gender Distribution

Female students slightly outnumber male students in the dataset. The gender distribution is relatively balanced and allows meaningful comparisons between male and female student performance.

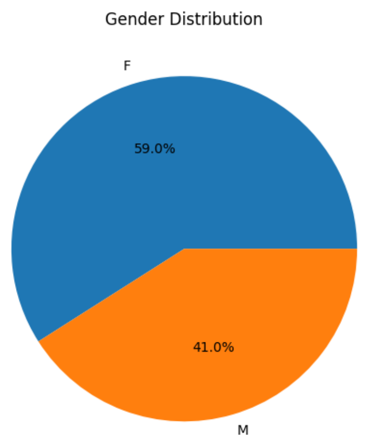

Figure 5.2: Gender Distribution

5.1.3 Urban vs Rural Student Distribution

Most students come from urban areas rather than rural regions. Urban students generally have better access to educational facilities and academic resources.

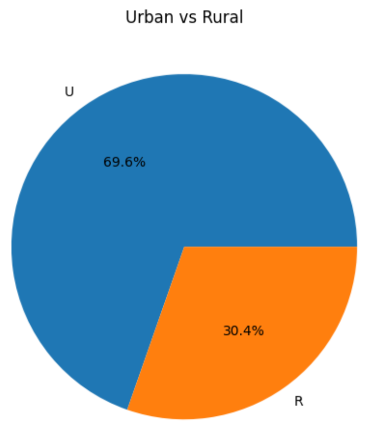

Figure 5.3: Urban vs Rural Student Distribution

5.1.4 Internet Access at Home

A large percentage of students have internet access at home. Internet availability supports self-learning and provides access to educational resources.

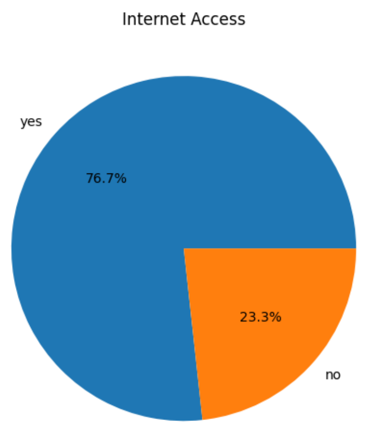

Figure 5.4: Internet Access at Home

5.1.5 Previous Academic Failures

Most students have no history of academic failures. Students without previous failures generally demonstrate stronger academic foundations.

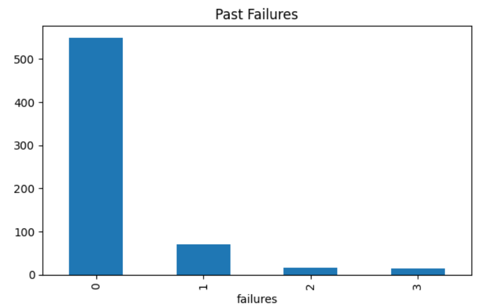

Figure 5.5: Previous Academic Failures

5.1.6 Study Time Distribution

Most students spend a moderate amount of time studying each week. Regular study habits contribute significantly to academic achievement.

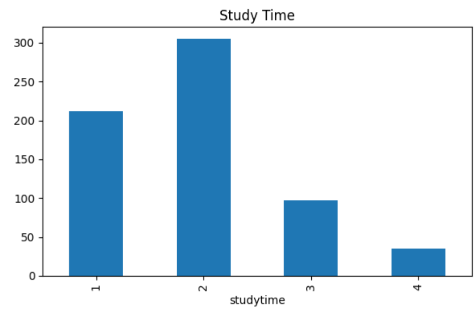

Figure 5.6: Study Time Distribution

5.1.7 Age Distribution

Students aged 16–18 form the largest group in the dataset. This age group represents the majority of secondary school students.

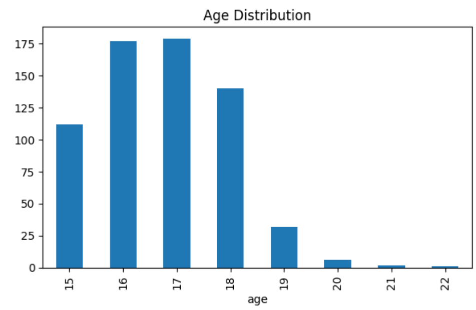

Figure 5.7: Age Distribution

5.2 Data Relationships

Relationship analysis was conducted to understand how different factors influence student performance.

5.2.1 Average Final Grade by Gender

Female students achieve slightly higher average final grades than male students. The difference is small but noticeable.

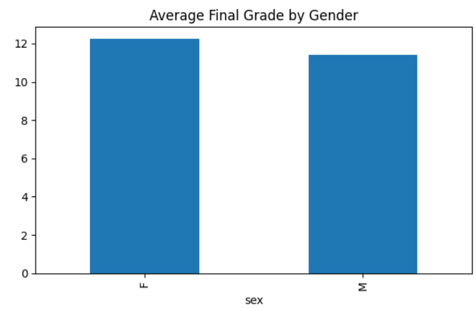

Figure 5.8: Average Final Grade by Gender

5.2.2 Average Final Grade by Residence Area

Urban students perform slightly better than rural students, possibly due to better access to educational infrastructure and learning resources.

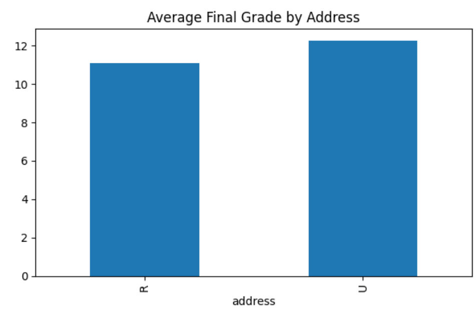

Figure 5.9: Average Final Grade by Residence Area

5.2.3 Average Final Grade by Number of Failures

Academic performance decreases as the number of previous failures increases. This is one of the strongest trends observed in the dataset.

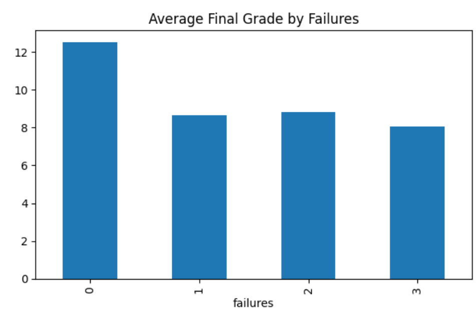

Figure 5.10: Average Final Grade by Number of Failures

5.3 Key Metrics Identification

Several key metrics were identified during the analysis:

- Average Final Grade (G3)
- Average Study Time
- Academic Failure Rate
- Internet Accessibility
- Gender Distribution
- Urban vs Rural Distribution
- Attendance and Absence Trends
- Family Educational Support

These metrics provide valuable insights into student performance and help educational institutions make data-driven decisions.